# Algebra Monsoon 2026 Lecture 1

**Def:** law of composition, magma $E$ on $E$

→ 0) $\emptyset$ and $\{e\}$ are magmas.

Example 1) $E = \mathcal{P}(X)$, $(A, B) \mapsto A \cup B$

2) $E = \mathcal{P}(X)$, $(A, B) \mapsto A \cap B$

3) $E = \mathbb{Z}$, $(a, b) \mapsto a + b$ (add.)

4) $E = \mathbb{Z}$, $(a, b) \mapsto ab$ (mult.)

5) $E = \mathbb{N}$ for addition 6) $E = \mathbb{N}$ for multiplication

7) all mappings $X \to X$ under $\circ$.

**Def Morphism** of magmas $f : M \to N$

$f(xy) = f(x)f(y)$ i.e. $M \times M \to M$, $f \times f$, $f$, $N \times N \to N$ commutes.

• endomorphism

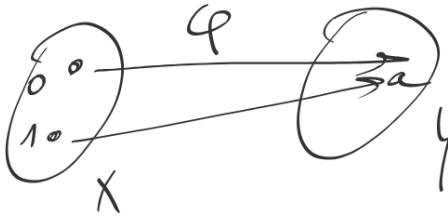
Ex. $x \mapsto 2x$ is a morphism of magmas $\mathbb{N} \to \mathbb{N}$ for addition, not for mult.

- Let $\varphi : X \to Y$ be a mapping.

$\varphi_* : \mathcal{P}(X) \to \mathcal{P}(Y)$ is a morphism for union, but not for intersection:

We have $\varphi_*(A \cup B) = \varphi_*(A) \cup \varphi_*(B)$

but the inclusion $\varphi_*(A \cap B) \subset \varphi_*(A) \cap \varphi_*(B)$ may be strict:




$\varphi_*(\{0\}) = \{a\}$

$\varphi_*(\{1\}) = \{a\}$

$\varphi_*(\underbrace{\{0\} \cap \{1\}}_{=\emptyset}) = \emptyset$

(+) (·) $B = \{0, 1\}$ with

| $\wedge$ | 0 | 1 |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |

| $\vee$ | 0 | 1 |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 1 |

and

(9) $\mathcal{B}(E)$ under $\Delta$

$A \Delta B = \complement_{A \cup B}(A \cap B)$.

So $\underbrace{\varphi_*(\{0\} \cap \{0\})}_{=\emptyset} \neq \underbrace{\varphi_*(\{0\}) \cap \varphi_*(\{1\})}_{=\{a\}}$

$= \phi$

**Proposition** 1) For every magma $M$, the identity mapping $Id_M : M \to M$ is a morphism of magmas.

2) If $f : M \to N$ and $g : N \to P$ are morphisms of magmas, then so is $g \circ f : M \to P$.

**Proof** 1) $Id_M(xy) = xy = Id_M(x)\, Id_M(y)$.

2) $$\begin{aligned}(g \circ f)(xy) &= g(f(xy)) \\ &= g(f(x) f(y)) \\ &= g(f(x))\, g(f(y)) \\ &= (g \circ f)(x)\, (g \circ f)(y).\end{aligned}$$

For every magma $M$, there are unique morphisms of magmas $\emptyset \longrightarrow M$ and $M \longrightarrow$ yes.

$f : M \to N$

magmas' $\varnothing \longrightarrow$ [illegible]

**Def** A morphism of magmas $f : M \longrightarrow N$ is an isomorphism if there exists a morphism of magmas $g : N \longrightarrow M$ such that

$$\begin{cases} g \circ f = Id_M \\ f \circ g = Id_N \end{cases}.$$

An endomorphism which is an isomorphism is called an automorphism.

**Proposition** If $g$ exists, it is unique.

Proof: Indeed, the existence of $g$ implies that $f$ is a bijection and $g = f^{-1}$. □

So an isomorphism of magmas is a bijective morphism. Conversely,

**Proposition** A bijective morphism of magmas is an isomorphism.

**Proof**: Let $f: M \to N$ be a bijective morphism of magmas. It is enough to show that the reciprocal bijection $f^{-1}: N \to M$ is a morphism of magmas. Now,

(1) $f^{-1}(ab) = f^{-1}(a)\, f^{-1}(b)$

is equivalent to

(2) $f\left(f^{-1}(ab)\right) = f\left(f^{-1}(a)\, f^{-1}(b)\right)$

because $f$ is a bijection. But $f$ is also a morphism of magmas, so

$$f\left(f^{-1}(a)\, f^{-1}(b)\right) = f\left(f^{-1}(a)\right) f\left(f^{-1}(b)\right) = ab$$

Clearly, $f\left(f^{-1}(ab)\right) = ab$, so (2) holds and therefore (1) holds. □

$a + a \in M$

and therefore $(*)$

**Def** Let $M$ be a magma. Two elements $x, y \in M$ are said to be **permutable** if $xy = yx$.

**Def** A magma $M$ is said to be **associative** if $(xy)z = x(yz)$ for all $x, y, z \in M$.

**Commutative** if $xy = yx$ for all $x, y \in M$.

Example $M = \mathrm{End}(X)$, the set of all mappings $X \to X$. is associative for the usual composition of mappings.

If $M$ is associative, we may write $xyz$ without ambiguity. More generally,

**Proposition** Let $M$ be a magma, $\varphi : M \times M \longrightarrow M$ its law of composition. If $M$ is associative, then there is a unique family of mappings

$$\varphi_n : M^n \longrightarrow M, \quad n \geq 1,$$

such that 1) $\varphi_1 = Id_M$

2) $\varphi_2 = \varphi$

and 3) $\varphi_n(a_1, a_2, \dots, a_n)$

$$= \varphi\left(\varphi_i(a_1, \dots, a_i), \varphi_{n-i}(a_{i+1}, \dots, a_n)\right)$$

for every $i \in [1, n-1]$.

Proof This is vacously true for $n = 1$.

we have $\varphi_2(a_1, a_2) = \varphi(a_1, a_2)$

$= \varphi(\varphi_1(a_1), \varphi_1(a_2))$

so the statement is true for $n = 2$.

Suppose that $\varphi_1, \varphi_2, \dots, \varphi_n$ have been uniquely defined for <u>some</u> $n \geq 2$.

If $\varphi_{n+1}$ exists, it must satisfy

$\varphi_{n+1}(a_1, \dots, a_{n+1}) = \varphi(\varphi_n(a_1, \dots, a_n), a_{n+1})$,

which is a special case of 3). It follows that $\varphi_{n+1}$ is unique (if it exists).

Let us <u>define</u> $\varphi_{n+1}$ by this formula and check that it satisfies 3). If $i = n$, then 3) holds by the definition of $\varphi_{n+1}$. If $i < n$, then

$\varphi_{n+1}$ ...

Let us define $\varphi_{n+1}$ by this formula and check that it satisfies 3). If $i = n$, then 3) holds by the definition of $\varphi_{n+1}$. If $i < n$, then

$$\varphi_{n+1}(a_1, \dots, a_{n+1})$$

$$= \varphi(\varphi_n(a_1, \dots, a_n), a_{n+1}) \quad \text{(def)}$$

$$= \varphi(\varphi(\varphi_i(a_1, \dots, a_i), \varphi_{n-i}(a_{i+1}, \dots, a_n)), a_{n+1})$$

(by 3))

$$= \varphi(\varphi_i(a_1, \dots, a_i), \varphi(\varphi_{n-i}(a_{i+1}, \dots, a_n), a_{n+1}))$$

(by associativity)

$$= \varphi(\varphi_i(a_1, \dots, a_i), \varphi_{n+1-i}(a_{i+1}, \dots, a_{n+1}))$$

(by 3))

showing that 3) holds for $\varphi_{n+1}$ and completing the proof. □

... without ambiguity

computing

· We write $x_1 x_2 \cdots x_n$ for $\varphi_n(x_1, x_2, \ldots, x_n)$ without ambiguity.

Let $M$ be a magma. For every $a \in M$, we have the mappings

$$\gamma_a, \delta_a : M \to M$$

defined by $\gamma_a(x) = ax$ (left translation by $a$)

$\delta_a(x) = xa$ (right translation by $a$)

Def. We say that $a$ is left cancellable if $\gamma_a$ is injective ($ax = ay \Rightarrow x = y$).

We say that $a$ is right cancellable if $\delta_a$ is injective.

**Def** Let $M$ be a magma. An element $e \in M$ is said to be a *left unit* if

$$ex = x \quad \text{for every } x \in M.$$

a *right unit* if $xe = x$ for all $x \in M$.

Equivalently $\gamma_e = Id$, resp. $\delta_e = Id$.

**Proposition** If a magma has a left unit $e$ and a right unit $f$, then $e = f$.

**Proof** We have $ef = e$ because $e$ is a left unit and $ef = f$ because $f$ is a right unit.

Hence $e = f$. □

**Def** Let $M$ be a magma. An element $e \in M$ is called a unit (or a neutral element) if it is both a left unit and a right unit.

If a unit exists, it is unique and called the unit.

**Def** A magma is said to be unital if it has a neutral element.

- $\{e\}$ is unital but $\emptyset$ is not.

**Ex.** In the magma $(\mathcal{P}(X), \cup)$, the empty set $\emptyset$ is a neutral element.

In $(\mathcal{P}(X), \cap)$, the neutral element is $X$.

In $(\mathbb{N}, +)$ the neutral element is $0$.

In $(\mathbb{N}, \times)$ ——————— $1$.

**Def** Let $M$ be a unital magma, $e$ its neutral element. An element $x \in M$ is said to be left invertible if there exists $y \in M$ such that $yx = e$. Such a $y$ is called a left inverse of $x$.

**Def** $x$ is right invertible $\Leftrightarrow e \in r_{x*}(M)$.

**Def** An associative unital magma is called a monoid.

Ex. For any set $E$, the magma $\mathrm{End}(E)$ is a monoid.

Example: $\mathbb{N}$ is a monoid under addition.

**Proposition** Let $M$ be a monoid. If $x_1, x_2 \in M$ are left invertible, then so is $x_1 x_2$.

If $y_1, y_2$ are left inverses of $x_1, x_2$ then $y_2 y_1$ is a left inverse of $x_1 x_2$.

Proof $(y_2 y_1)(x_1 x_2)$

$= y_2 (y_1 x_1) x_2$

$= y_2 e x_2$

$= y_2 x_2$

$= e.$ □

Proposition Let $M$ be a monoid. If $x \in M$ is left invertible, then it is left cancellable.

Proof. Let $y$ be a left inverse of $x$.

We have $xa = xb$

$\Rightarrow y(xa) = y(xb)$

$\Rightarrow (yx)a = (yx)b$

$\Rightarrow ea = eb$

$\Rightarrow a = b.$ □

then $y = z$.

Proof We have $yx = e$ and $xz = e$.

Therefore $y = ey$

$= (yx)y$

$= y(xy)$

$= ye$

$= y$.

□

**Def** Let $M$ be a monoid. An element $x \in M$ is said to be *invertible* if it has a left inverse and a right inverse.

Example: $e$ is both left and right invertible

If $x$ is invertible, then there is a unique $y \in M$ such that $yx = e$ (or $xy = e$).

(for if $y_1 x = e$, $y_2 x = e$, $xz = e$, then $y_1 = z$, $y_2 = z$, so $y_1 = y_2$).

This $y$ is denoted $x^{-1}$. Example: $e^{-1} = e$.

We have $x x^{-1} = e$

- If $x$ is invertible, then so is $x^{-1}$ and $(x^{-1})^{-1} = x$.
- If $x, y$ are invertible, then so is $xy$ and $(xy)^{-1} = y^{-1} x^{-1}$.

Notation: $M^\times$ is the set of invertible elements of $M$.
Ex. If $M$ is the mult. monoid $\mathbb{Z}$, then $\mathbb{Z}^\times = \{+1, -1\}$.

**Def** Let $M, N$ be monoids. A morphism of monoids $f: M \to N$ is a mapping such that 1) $f$ is a morphism of magmas ($f(xy) = f(x)f(y)$)

and 2) $f(e_M) = e_N$

There are unique morphisms of monoids $\{e\} \to M$ and $M \to \{e\}$.

- Id is a morphism
- $M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3$ are morphisms, then so is $f_2 \circ f_1$

. $\mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$, $x \mapsto (x, 0)$ is not a morphism of malt. monoids because $1 \mapsto (1,0)$, not $(1,1)$.

**Proposition** Let $M$ be a monoid and $x \in M$. There is a unique morphism of monoids $j: \mathbb{N} \to M$ such that $j(1) = x$

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Proof Uniqueness: Let $f_1, f_2$ be two such morphisms. We prove by induction on $n$ that $f_1(n) = f_2(n)$ for every $n \in \mathbb{N}$.

By definition, $f_1(0) = e$ and $f_2(0) = e$,

and by hypothesis $f_1(1) = x$ and $f_2(1) = x$.

Suppose we have $f_1(n) = f_2(n)$ for some $n \geq 1$.

Then $f_1(n+1) = f_1(n) f_1(1)$

$= f_1(n) x$

and similarly $f_2(n+1) = f_2(n) x$;

it follows that $f_1(n+1) = f_2(n+1)$

Existence: Define $j$ inductively by

$$j(n+1) = j(n)\,x$$

We prove that $j(n+m) = j(n)\,j(m)$ by induction on $m$. The case $m = 1$ is true by definition. If this identity holds for <u>some</u> $m \geq 1$, then

$$\begin{aligned} j(n+(m+1)) &= j((n+m)+1) \\ &= j(n+m)\,x \\ &= j(n)\,j(m)\,x \\ &= j(n)\,j(m+1) \end{aligned}$$

If $x = e$, then $j$ is the constant mapping $e$.

If $M = N$, then $j(n) = n\,x$. ("multiples of $x$") □

**Def** Let $M$ be a monoid, $x \in M$, and $f: \mathbb{N} \to M$ the unique morphism of monoids such that $f(1) = x$.
We write $x^n = f(n)$ for $n \in \mathbb{N}$.

We have $x^0 = e$

$x^{m+n} = x^m x^n$ for $m, n \in \mathbb{N}$.

$(x^m)^n = x^{mn}$

**Proposition** Let $\varphi : M \to N$ be a morphism of monoids. If $x \in M$ is invertible, then $\varphi(x) \in N$ is invertible and $\varphi(x)^{-1} = \varphi(x^{-1})$.

**Proof** We have

$$\begin{aligned} e_N &= \varphi(e_M) \\ &= \varphi(x x^{-1}) \\ &= \varphi(x)\varphi(x^{-1}), \end{aligned}$$

so $\varphi(x^{-1})$ is a right inverse $\varphi(x)$.

Similarly, $\varphi(x^{-1})$ is a left inverse of $\varphi(x)$

Hence $\varphi(x^{-1}) = \varphi(x)^{-1}$.

□

**Def** A group is a monoid in which every element is invertible.

Let $G, H$ be groups. A morphism of groups $\varphi : G \to H$ is a mapping such that

1) $\varphi$ is a morphism of monoids.
2) $\varphi(x^{-1}) = \varphi(x)^{-1}$ for every $x \in G$.

**Example** $\mathbb{Z}$ is a (commutative) group under addition.

$\mathbb{Z}$ is not a group under multiplication, only a (commutative) monoid.

**Proposition** Let $G, H$ be groups.
Every morphism of magmas $\varphi : G \to H$
is a morphism of groups.

**Proof** We are given that

$$\varphi(xy) = \varphi(x)\varphi(y) \quad \text{for all } x, y \in G$$

and we have to prove that

$$\varphi(e_G) = e_H$$

and $\varphi(x^{-1}) = \varphi(x)^{-1}$ for every $x \in G$.

From $e_G e_G = e_G$, it follows that

$$\varphi(e_G)\varphi(e_G) = \varphi(e_G).$$

Multiplying throughout by $\varphi(e_G)^{-1}$, we get

$$\varphi(e_G) = e_H;$$

this means that $\varphi$ is a morphism of monoids;
as the identity $\varphi(x^{-1}) = \varphi(x)^{-1}$ was
proved earlier, we are done. □

- For every group $G$, $\mathrm{Id}_G : G \to G$ is a morphism of groups.

- If $f : G \to H$ and $g : H \to K$ are morphisms of groups, then so is $g \circ f : G \to K$.

- A bijective morphism of groups $\varphi$ is an isomorphism of groups.

Proof: We have seen that $\varphi^{-1}$ is a morphism of magmas hence a morphism of groups.

**Proposition** Let $M$ be a monoid, $x \in M$, and $j : \mathbb{N} \to M$ the unique morphism of monoids such that $j(1) = x$. If $x$ is invertible in $M$, there is a unique morphism of monoids $\tilde{j} : \mathbb{Z} \to M$ extending $j$ (meaning $\tilde{j}|_{\mathbb{N}} = j$)

**Proof** Suppose that $x$ is invertible in $M$. We have seen that there is a unique morphism of monoids $k : \mathbb{N} \to M$ such that $k(1) = x^{-1}$. We have $k(0) = j(0)$.

Define $\tilde{j}(n) = \begin{cases} j(n) & \text{if } n \in \mathbb{N} \\ k(-n) & \text{if } n \notin \mathbb{N}. \end{cases}$

(so that $\tilde{j}(-1) = k(1) = x^{-1}$, and $\tilde{j}(-n) = (x^{-1})^n$ for every $n > 0$)

□

We write $\tilde{f}(n) = x^n$.

The notation is consistent

- for $n \in \mathbb{N}$, the new def is the same as the old one for monoids: $\tilde{f}(n) = f(n)$.
- for $n = -1$, $\tilde{f}(-1)$ is the inverse of $x$. (which is already denoted as $x^{-1}$).

The law of exponents

$$x^{m+n} = x^m x^n \quad , \quad (x^m)^n = x^{mn}$$

holds for all $m, n \in \mathbb{Z}$ (because $\tilde{f}$ is a morphism of monoids)

**Proposition** Let $G$ be a group and $x \in G$.
There is a unique morphism of groups
$\tilde{f} : \mathbb{Z} \to G$ such that $\tilde{f}(1) = x$.

**Proof** : This follows from the previous proposition because $x$ is invertible in $G$. □

For monoids $M_1, M_2$, denote by $\mathrm{Hom}(M_1, M_2)$ the set of morphisms of monoids $M_1 \to M_2$.

Exercise: Let $M$ be a monoid. Show that the mapping

$$\mathrm{Hom}(\mathbb{N}, M) \longrightarrow M, \quad f \mapsto f(1)$$

is a bijection.

Exercise: Let $G$ be a group. Show that the mapping

$$\mathrm{Hom}(\mathbb{Z}, G) \longrightarrow G, \quad f \mapsto f(1)$$

is a bijection.

Def A commutative group is called a module.